2022-Ph-H1-Q5

Section: Our Dynamic Universe
Topic: Forces, Energy and Power

Summary:
A 50.0 kg student stands on a newton balance inside a lift accelerating **upwards** at 1.2 m/s². The reading on the balance is required.

Solution:
The apparent weight is:
$R = m(g + a) = 50.0(9.8 + 1.2) = 50.0 \times 11.0 = 550\,\mathrm{N}.$

Answer: D. 550 N.

Guidance for Students:

- In accelerating lifts, **apparent weight** differs from actual weight.
- Upward acceleration increases the normal force (balance reading).

Revision Tips:

- $R = m(g + a)$ when accelerating upwards.
- If the lift accelerates **downwards**, use $R = m(g - a)$.